

04 — GAN 3D Normalized

Dự án: Latent Manipulation of Brain MRI using Volume-Preserving GANs

Input: NIfTI normalized từ 00b_preprocessing_3d.ipynb

Kiến trúc:

- **Generator:** 3D U-Net + Age Embedding inject vào bottleneck
- **Discriminator:** 3D PatchGAN + Age Conditioning

Output:

```
gan3d_normalized/  
└─ best_model.pth
```

Bước 1: Cấu hình

```
import os  
  
# ==== ĐƯỜNG DẪN ====  
DATA_DIR =  
'/kaggle/input/datasets/minhbodoi/3000-preprocessed-3d/preprocessed_3d/  
normalized'  
LABELS_CSV =  
'/kaggle/input/datasets/minhbodoi/3000-preprocessed-3d/preprocessed_3d/  
preprocessing_log.csv'  
OUTPUT_DIR = '/kaggle/working/conditional_gan3d_normalized'  
  
os.makedirs(OUTPUT_DIR, exist_ok=True)  
  
# ==== HYPERPARAMETERS ====  
VOLUME_SIZE = 64      # resize về 64×64×64 vì 3D nặng hơn 2D rất  
nhiều  
BATCH_SIZE = 1        # 3D nặng, chỉ dùng batch=1  
NUM_EPOCHS = 300  
PATIENCE = 20         # dừng nếu val_SSIM không tăng >= MIN_DELTA sau 20  
epoch liên tiếp  
LR_G = 2e-4  
LR_D = 2e-4  
LAMBDA_L1 = 10        # giảm từ 100 → 10  
LAMBDA_AGE = 5        # tăng từ 1 → 5  
LAMBDA_GP = 10        # gradient penalty WGAN-GP  
LATENT_DIM = 256  
  
# Tìm file .nii hoặc .nii.gz (Kaggle tự giải nén .nii.gz thành .nii)  
nii_files = [f for f in os.listdir(DATA_DIR)  
              if f.endswith('.nii.gz') or f.endswith('.nii')]
```

```
print(f'Data dir : {DATA_DIR}')
print(f'NII files: {len(nii_files)}')
```

```
Data dir :
/kaggle/input/datasets/minhbodoi/3000-preprocessed-3d/preprocessed_3d/
normalized
NII files: 3060
```

Bước 2: Import thư viện

```
!pip install nibabel -q
```

```
import torch
import torch.nn as nn
import torch.optim as optim
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader
import nibabel as nib
import numpy as np
import pandas as pd
from tqdm.notebook import tqdm
import warnings
warnings.filterwarnings('ignore')
```

```
DEVICE = 'cuda' if torch.cuda.is_available() else 'cpu'
print(f'Device: {DEVICE}')
if DEVICE == 'cuda':
    print(f'GPU : {torch.cuda.get_device_name(0)}')
    print(f'VRAM: {torch.cuda.get_device_properties(0).total_memory /
1e9:.1f} GB')
```

```
Device: cuda
GPU : Tesla T4
VRAM: 15.6 GB
```

Bước 3: Dataset

```
def find_nii(data_dir, subject_id):
    """Tìm file .nii hoặc .nii.gz – xử lý ca' 2 trường hợp."""
    for ext in ['.nii.gz', '.nii']:
        path = os.path.join(data_dir, f'{subject_id}{ext}')
        if os.path.exists(path):
            return path
    return None
```

```
class BrainMRI3DDataset(Dataset):
    def __init__(self, data_dir, labels_csv, volume_size=64):
        """
        Load file NIfTI 3D, resize về`
```

```

volume_size×volume_size×volume_size.
    Resize nhỏ để tiết kiệm RAM khi train 3D.
    """
    self.data_dir = data_dir
    self.volume_size = volume_size

    df = pd.read_csv(labels_csv)
    df['nii_path'] = df['subject_id'].apply(
        lambda x: find_nii(data_dir, x)
    )
    df = df[df['nii_path'].notna()].reset_index(drop=True)

    self.df = df
    self.age_min = df['age'].min()
    self.age_max = df['age'].max()

    print(f'Dataset: {len(df)} subjects | tuổi
    [{self.age_min:.1f}, {self.age_max:.1f}]')

    def normalize_age(self, age):
        return 2 * (age - self.age_min) / (self.age_max -
self.age_min) - 1

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        row = self.df.iloc[idx]
        data =
nib.load(row['nii_path']).get_fdata().astype(np.float32)

        # Resize về volume_size
        vol = torch.tensor(data).unsqueeze(0).unsqueeze(0) #
(1,1,H,W,D)
        vol = F.interpolate(vol, size=(self.volume_size,) * 3,
                             mode='trilinear', align_corners=False)
        vol = vol.squeeze(0) # (1, V, V, V)

        # Normalize về [-1, 1]
        vol = vol * 2 - 1

        age_norm = torch.tensor(self.normalize_age(row['age']),
dtype=torch.float32)
        age_raw = torch.tensor(row['age'] / 100.0,
dtype=torch.float32)
        return vol, age_norm, age_raw

dataset = BrainMRI3DDataset(DATA_DIR, LABELS_CSV, VOLUME_SIZE)

```

```
# — Train / Test split 80/20 —————
from torch.utils.data import random_split

full_dataset = BrainMRI3DDataset(DATA_DIR, LABELS_CSV, VOLUME_SIZE)
n_total = len(full_dataset)
n_train = int(0.8 * n_total)
n_test = n_total - n_train

train_set, test_set = random_split(
    full_dataset, [n_train, n_test],
    generator=torch.Generator().manual_seed(42)
)

dataset = full_dataset # đề' lấy age_min/max
dataloader = DataLoader(train_set, batch_size=BATCH_SIZE,
                        shuffle=True, num_workers=2,
                        pin_memory=True)
dataloader_test = DataLoader(test_set, batch_size=1,
                             shuffle=False, num_workers=2,
                             pin_memory=True)

print(f'Train: {n_train} subjects | Test: {n_test} subjects')

Dataset: 3060 subjects | tuổi [18.0, 88.0]
Dataset: 3060 subjects | tuổi [18.0, 88.0]
Train: 2448 subjects | Test: 612 subjects
```

Bước 4: Kiến trúc Model 3D

Giống file 01 nhưng thay Conv2d → Conv3d, ConvTranspose2d → ConvTranspose3d.

```
class AgeEmbedding(nn.Module):
    def __init__(self, embed_dim=256):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(1, 128), nn.ReLU(),
            nn.Linear(128, embed_dim)
        )
    def forward(self, age):
        return self.net(age.unsqueeze(-1))

class UNetBlock3D(nn.Module):
    """Block 3D: Conv3d thay vì Conv2d."""
    def __init__(self, in_ch, out_ch, down=True, use_bn=True,
                 dropout=False):
        super().__init__()
        layers = []
        if down: layers.append(nn.Conv3d(in_ch, out_ch, 4, 2, 1,
```

```

bias=False))
    else : layers.append(nn.ConvTranspose3d(in_ch, out_ch, 4,
2, 1, bias=False))
    if use_bn : layers.append(nn.BatchNorm3d(out_ch))
    if dropout : layers.append(nn.Dropout(0.5))
    layers.append(nn.LeakyReLU(0.2) if down else nn.ReLU())
    self.block = nn.Sequential(*layers)
def forward(self, x):
    return self.block(x)

class Generator3D(nn.Module):
    """
    3D U-Net Generator với age conditioning.
    Input : volume (B, 1, 64, 64, 64) + tuổi (B,)
    Output: volume sinh ra (B, 1, 64, 64, 64)
    """
    def __init__(self, latent_dim=256):
        super().__init__()
        self.age_embed = AgeEmbedding(latent_dim)
        self.age_proj = nn.Linear(latent_dim, 256)
        self.age_fuse = nn.Conv3d(512, 256, 1) # concat e4(256) +
age(256) → 256

        # Encoder
        self.e1 = UNetBlock3D(1, 32, down=True, use_bn=False) # 32
        self.e2 = UNetBlock3D(32, 64, down=True) #
16
        self.e3 = UNetBlock3D(64, 128, down=True) #
8
        self.e4 = UNetBlock3D(128, 256, down=True, use_bn=False) #
4 bottleneck

        # Decoder
        self.d1 = UNetBlock3D(256, 128, down=False, dropout=True) # 8
        self.d2 = UNetBlock3D(256, 64, down=False) #
16
        self.d3 = UNetBlock3D(128, 32, down=False) #
32

        self.out = nn.Sequential(
            nn.ConvTranspose3d(64, 1, 4, 2, 1),
            nn.Tanh()
        )

    def forward(self, x, age):
        e1 = self.e1(x)
        e2 = self.e2(e1)
        e3 = self.e3(e2)
        e4 = self.e4(e3)

```

```

        age_feat = self.age_proj(self.age_embed(age)).view(-1, 256, 1,
1, 1).expand_as(e4)
        z = self.age_fuse(torch.cat([e4, age_feat], dim=1)) # concat
→ model tự học balance
        d1 = self.d1(z)
        d2 = self.d2(torch.cat([d1, e3], dim=1))
        d3 = self.d3(torch.cat([d2, e2], dim=1))
        return self.out(torch.cat([d3, e1], dim=1))

class Discriminator3D(nn.Module):
    """
    3D PatchGAN Discriminator với age conditioning.
    Input : volume (B, 1, 64, 64, 64) + age channel → (B, 2, 64, 64,
64)
    """
    def __init__(self):
        super().__init__()
        self.model = nn.Sequential(
            nn.Conv3d(2, 32, 4, 2, 1, bias=False),
nn.LeakyReLU(0.2),
            nn.Conv3d(32, 64, 4, 2, 1, bias=False),
nn.BatchNorm3d(64), nn.LeakyReLU(0.2),
            nn.Conv3d(64, 128, 4, 2, 1, bias=False),
nn.BatchNorm3d(128), nn.LeakyReLU(0.2),
            nn.Conv3d(128, 1, 4, 1, 1)
        )
    def forward(self, vol, age):
        age_map = age.view(-1, 1, 1, 1, 1).expand(
            -1, 1, vol.shape[2], vol.shape[3], vol.shape[4]
        )
        return self.model(torch.cat([vol, age_map], dim=1))

G = Generator3D(LATENT_DIM).to(DEVICE)
D = Discriminator3D().to(DEVICE)
print(f'Generator3D : {sum(p.numel() for p in
G.parameters())/1e6:.1f}M params')
print(f'Discriminator3D: {sum(p.numel() for p in
D.parameters())/1e6:.1f}M params')

Generator3D : 6.4M params
Discriminator3D: 0.7M params

```

Bước 5: Loss + Optimizers

```

# — WGAN-GP —————
criterion_l1 = nn.L1Loss()
criterion_age = nn.MSELoss()

```

```

def compute_gradient_penalty_3d(D, real, fake, ages, device):
    B = real.size(0)
    alpha = torch.rand(B, 1, 1, 1, 1, device=device)
    interp = (alpha * real + (1 - alpha) * fake).requires_grad_(True)
    d_interp = D(interp, ages)
    grad = torch.autograd.grad(
        outputs=d_interp, inputs=interp,
        grad_outputs=torch.ones_like(d_interp),
        create_graph=True, retain_graph=True
    )[0]
    gp = ((grad.norm(2, dim=1) - 1) ** 2).mean()
    return gp

age_regressor = nn.Sequential(
    nn.AdaptiveAvgPool3d(4), nn.Flatten(),
    nn.Linear(64, 64), nn.ReLU(),
    nn.Linear(64, 1), nn.Sigmoid()
).to(DEVICE)

opt_G = optim.Adam(
    list(G.parameters()) + list(age_regressor.parameters()),
    lr=LR_G, betas=(0.0, 0.9)
)
opt_D = optim.Adam(D.parameters(), lr=LR_D, betas=(0.0, 0.9))

scheduler_G = optim.lr_scheduler.StepLR(opt_G, step_size=50,
gamma=0.5)
scheduler_D = optim.lr_scheduler.StepLR(opt_D, step_size=50,
gamma=0.5)

print('Loss + Optimizers (WGAN-GP 3D) sẵn sàng!')
Loss + Optimizers (WGAN-GP 3D) sẵn sàng!

```

Bước 6: Training

```

from skimage.metrics import structural_similarity as ssim_fn
import numpy as np, subprocess, shutil, tempfile

# — Duong dan checkpoint —————
CKPT_PATH      = f'{OUTPUT_DIR}/last_checkpoint.pth'
BEST_PATH      = f'{OUTPUT_DIR}/best_model.pth'

# — Kaggle Dataset info —————
_ku = subprocess.run('kaggle config view', shell=True,
capture_output=True, text=True).stdout
KAGGLE_USER    = [l.split(':')[1].strip() for l in _ku.split('\n') if
'username' in l][0]
DATASET_NAME   = os.path.basename(OUTPUT_DIR).replace('_', '-')
MIN_DELTA      = 0.005 # chỉ tính là cải thiện khi SSIM tăng >=

```

0.005

```
def push_checkpoints_to_kaggle(msg):
    tmp = tempfile.mkdtemp()
    try:
        for fn in ['last_checkpoint.pth', 'best_model.pth']:
            if os.path.exists(f'{OUTPUT_DIR}/{fn}'):
                shutil.copy2(f'{OUTPUT_DIR}/{fn}', f'{tmp}/{fn}')
        import json as _j
        with open(f'{tmp}/dataset-metadata.json', 'w') as _f:
            _j.dump({'title': DATASET_NAME,
                    'id' : f'{KAGGLE_USER}/{DATASET_NAME}',
                    'licenses': [{ 'name': 'CC0-1.0' }]}, _f)
        chk = subprocess.run(
            f'kaggle datasets list --user {KAGGLE_USER} --search {DATASET_NAME}',
            shell=True, capture_output=True, text=True)
        if DATASET_NAME in chk.stdout:
            subprocess.run(f'kaggle datasets version -p {tmp} -m "{msg}"', shell=True)
        else:
            subprocess.run(f'kaggle datasets create -p {tmp}',
            shell=True)
        print(f' Cloud: {msg} -> {KAGGLE_USER}/{DATASET_NAME}')
    finally:
        shutil.rmtree(tmp)

# — Resume neu co checkpoint —————
start_epoch = 1
patience_count = 0
best_val_ssim = -1.0
history = {'loss_G': [], 'loss_D': [], 'loss_L1': [], 'loss_age': [],
          'val_ssim': []}

def find_checkpoint(filename):
    p = f'{OUTPUT_DIR}/{filename}'
    if os.path.exists(p): return p, 'working'
    import glob
    matches = glob.glob(f'/kaggle/input/**/{filename}',
    recursive=True)
    if matches: return matches[0], 'input'
    return None, None

load_path = None
for fname in ['last_checkpoint.pth', 'best_model.pth']:
    p, src_loc = find_checkpoint(fname)
    if p:
        load_path = p
        print(f'Tim thay {fname} ({src_loc}) — load de train tiep...')
        break
```



```

if not load_path:
    print('Khong co checkpoint - train tu dau')

if load_path:
    ckpt = torch.load(load_path, map_location=DEVICE)
    G.load_state_dict(ckpt['G_state'])
    D.load_state_dict(ckpt['D_state'])
    opt_G.load_state_dict(ckpt['opt_G'])
    opt_D.load_state_dict(ckpt['opt_D'])
    start_epoch = ckpt['epoch'] + 1
    best_val_ssim = ckpt.get('best_val_ssim', -1.0)
    patience_count = ckpt.get('patience_count', 0)
    history = ckpt.get('history', history)
    print(f'Resume tu epoch {ckpt["epoch"]} |
best_val_SSIM={best_val_ssim:.4f}')
    print(f'    Tiep tuc tu epoch {start_epoch} -> {NUM_EPOCHS}')

```

```

        real_vols, ages_norm, ages_raw = next(data_iter)
    except StopIteration:
        data_iter = iter(dataloader)
        real_vols, ages_norm, ages_raw = next(data_iter)
    real_vols = real_vols.to(DEVICE)
    ages_norm = ages_norm.to(DEVICE)
    ages_raw = ages_raw.to(DEVICE)
    opt_G.zero_grad()
    fake_vols = G(real_vols, ages_norm)
    loss_G_adv = -D(fake_vols, ages_norm).mean()
    loss_L1 = criterion_l1(fake_vols, real_vols) * LAMBDA_L1
    pred_age = age_regressor(fake_vols).squeeze()
    loss_age = criterion_age(pred_age, ages_raw) * LAMBDA_AGE
    loss_G = loss_G_adv + loss_L1 + loss_age
    loss_G.backward()
    opt_G.step()
    epoch_loss_G += loss_G_adv.item()
    epoch_loss_L1 += loss_L1.item()
    epoch_loss_age += loss_age.item()
    n_batches += 1

scheduler_G.step()
scheduler_D.step()

n = max(n_batches, 1)
avg_loss_G = epoch_loss_G / n
avg_loss_D = epoch_loss_D / (n * N_CRITIC)
avg_loss_L1 = epoch_loss_L1 / n
avg_loss_age = epoch_loss_age / n

# — SSIM tren test set voi shuffled age —————
G.eval()
ssim_scores = []
with torch.no_grad():
    all_vols, all_ages = [], []
    for real_vols, ages_norm, _ in dataloader_test:
        all_vols.append(real_vols)
        all_ages.append(ages_norm)
    all_vols = torch.cat(all_vols, dim=0)
    all_ages = torch.cat(all_ages, dim=0)
    shuffled_idx = torch.randperm(len(all_ages))
    all_ages_shuf = all_ages[shuffled_idx]
    for i in range(len(all_vols)):
        vol = all_vols[i:i+1].to(DEVICE)
        age = all_ages_shuf[i:i+1].to(DEVICE)
        fake = G(vol, age)
        r = (vol[0, 0].cpu().numpy() + 1) / 2
        f = (fake[0, 0].cpu().numpy() + 1) / 2
        slice_scores = [ssim_fn(r[j], f[j], data_range=1.0)
                        for j in range(r.shape[0])]

```

```

        ssim_scores.append(float(np.mean(slice_scores)))
val_ssim = float(np.mean(ssim_scores))

history['loss_G'].append(avg_loss_G)
history['loss_D'].append(avg_loss_D)
history['loss_L1'].append(avg_loss_L1)
history['loss_age'].append(avg_loss_age)
history['val_ssim'].append(val_ssim)

# — Luu last_checkpoint —————
torch.save({
    'epoch'          : epoch,
    'G_state'        : G.state_dict(),
    'D_state'        : D.state_dict(),
    'opt_G'          : opt_G.state_dict(),
    'opt_D'          : opt_D.state_dict(),
    'history'        : history,
    'age_min'        : dataset.age_min,
    'age_max'        : dataset.age_max,
    'best_val_ssim'   : best_val_ssim,
    'patience_count' : patience_count,
    'test_indices'    : test_set.indices,
}, CKPT_PATH)

# — Luu best model neu cai thien —————
if val_ssim > best_val_ssim + MIN_DELTA:
    best_val_ssim = val_ssim
    patience_count = 0
    torch.save({
        'epoch'          : epoch,
        'G_state'        : G.state_dict(),
        'D_state'        : D.state_dict(),
        'opt_G'          : opt_G.state_dict(),
        'opt_D'          : opt_D.state_dict(),
        'history'        : history,
        'age_min'        : dataset.age_min,
        'age_max'        : dataset.age_max,
        'best_loss_G'     : avg_loss_G,
        'best_val_ssim'   : best_val_ssim,
        'val_ssim'        : val_ssim,
        'test_indices'    : test_set.indices,
    }, BEST_PATH)
    print(f'Epoch {epoch:3d}/{NUM_EPOCHS} | '
          f'loss_G={avg_loss_G:.4f} | loss_D={avg_loss_D:.4f} | '
          f'loss_L1={avg_loss_L1:.4f} | val_SSIM={val_ssim:.4f} | '
          [BEST]')
else:
    patience_count += 1
    print(f'Epoch {epoch:3d}/{NUM_EPOCHS} | '
          f'loss_G={avg_loss_G:.4f} | loss_D={avg_loss_D:.4f} | '

```

```

        f'loss_L1={avg_loss_L1:.4f} | val_SSIM={val_ssim:.4f} |
        f'no improve {patience_count}/{PATIENCE}')

# — Push moi epoch —————
push_checkpoints_to_kaggle(f'{epoch}/{NUM_EPOCHS}')

# — Early stopping —————
if patience_count >= PATIENCE:
    print(f'Early stopping tai epoch {epoch}')
    break

print(f'\n=== TRAINING HOAN THANH ===')
print(f'Best val_SSIM: {best_val_ssim:.4f}')
print(f'Model luu tai: {OUTPUT_DIR}/best_model.pth')

```

Khong co checkpoint – train tu dau
 Bat dau training 300 epochs (WGAN-GP 3D)...

```
Epoch 1/300 | loss_G=-48.0504 | loss_D=-24.2023 | loss_L1=1.0223 |
val_SSIM=0.8911 | [BEST]
```

Warning: Looks like you're using an outdated `kaggle` version
 (installed: 2.0.1), please consider upgrading to the latest version
 (2.0.2)

Starting upload for file best_model.pth

```
100%|██████████| 81.0M/81.0M [00:01<00:00, 71.7MB/s]
19%|███████| 15.0M/81.0M [00:00<00:00, 144MB/s]
```

Upload successful: best_model.pth (81MB)
 Starting upload for file last_checkpoint.pth

```
100%|██████████| 81.0M/81.0M [00:01<00:00, 68.7MB/s]
```

Upload successful: last_checkpoint.pth (81MB)
 Your private Dataset is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsd/sds/conditional-gan3d-normalized>

```
Cloud: 1/300 -> cminhnguyndsd/sds/conditional-gan3d-normalized
Epoch 2/300 | loss_G=-109.1919 | loss_D=-8.4683 | loss_L1=0.4630 |
val_SSIM=0.7962 | no improve 1/20
```

Warning: Looks like you're using an outdated `kaggle` version
 (installed: 2.0.1), please consider upgrading to the latest version
 (2.0.2)

Starting upload for file best_model.pth

```
100%|██████████| 81.0M/81.0M [00:00<00:00, 85.8MB/s]
24%|███████| 19.5M/81.0M [00:00<00:00, 191MB/s]
```

Upload successful: best_model.pth (81MB)
 Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 62.5MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan3d-normalized>

Cloud: 2/300 -> cminhnguyndsdsds/conditional-gan3d-normalized
Epoch 3/300 | loss_G=-93.2571 | loss_D=-3.5513 | loss_L1=0.4061 |
val_SSIM=0.9301 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 66.9MB/s]
28%|███████| 22.8M/81.0M [00:00<00:00, 239MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 75.0MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan3d-normalized>

Cloud: 3/300 -> cminhnguyndsdsds/conditional-gan3d-normalized
Epoch 4/300 | loss_G=-84.0938 | loss_D=-2.2781 | loss_L1=0.3759 |
val_SSIM=0.9453 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 85.2MB/s]
27%|███████| 21.7M/81.0M [00:00<00:00, 225MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 75.5MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan3d-normalized>

Cloud: 4/300 -> cminhnguyndsdsds/conditional-gan3d-normalized
Epoch 5/300 | loss_G=-72.8891 | loss_D=-2.0947 | loss_L1=0.3487 |
val_SSIM=0.9519 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version

(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 79.6MB/s]
26%|███████| 21.0M/81.0M [00:00<00:00, 221MB/s]

Upload successful: best_model.pth (81MB)

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100%|██████████| 81.0M/81.0M [00:01<00:00, 61.5MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan3d-normalized>

Cloud: 5/300 -> cminhnguyndsdlds/conditional-gan3d-normalized
Epoch 6/300 | loss_G=-49.9277 | loss_D=-1.9696 | loss_L1=0.3193 |
val_SSIM=0.9568 | no improve 1/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 78.7MB/s]
0%| | 0.00/81.0M [00:00<?, ?B/s]

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100%|██████████| 81.0M/81.0M [00:00<00:00, 89.7MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan3d-normalized>

Cloud: 6/300 -> cminhnguyndsdlds/conditional-gan3d-normalized
Epoch 7/300 | loss_G=-21.9403 | loss_D=-1.6589 | loss_L1=0.2957 |
val_SSIM=0.9608 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 74.3MB/s]
28%|███████| 22.9M/81.0M [00:00<00:00, 238MB/s]

Upload successful: best_model.pth (81MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 81.3MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan3d-normalized>

Cloud: 7/300 -> cminhnguyndsdsds/conditional-gan3d-normalized
Epoch 8/300 | loss_G=7.1646 | loss_D=-1.3018 | loss_L1=0.2780 |
val_SSIM=0.9642 | no improve 1/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 92.9MB/s]
0%| | 0.00/81.0M [00:00<?, ?B/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 71.0MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan3d-normalized>

Cloud: 8/300 -> cminhnguyndsdsds/conditional-gan3d-normalized
Epoch 9/300 | loss_G=39.1458 | loss_D=-1.0513 | loss_L1=0.2570 |
val_SSIM=0.9672 | [BEST]
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 83.1MB/s]
32%|███████| 25.7M/81.0M [00:00<00:00, 270MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 83.4MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan3d-normalized>

Cloud: 9/300 -> cminhnguyndsdsds/conditional-gan3d-normalized
Epoch 10/300 | loss_G=55.0212 | loss_D=-0.8504 | loss_L1=0.2422 |
val_SSIM=0.9706 | no improve 1/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 64.2MB/s]
23%|███████| 18.2M/81.0M [00:00<00:00, 188MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 73.5MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 10/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 11/300 | loss_G=69.1731 | loss_D=-0.6790 | loss_L1=0.2341 |
val_SSIM=0.9694 | no improve 2/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 75.9MB/s]
26%|███████| 21.4M/81.0M [00:00<00:00, 215MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 72.4MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 11/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 12/300 | loss_G=76.5778 | loss_D=-0.5351 | loss_L1=0.2275 |
val_SSIM=0.9414 | no improve 3/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 92.7MB/s]
37%|███████| 30.0M/81.0M [00:00<00:00, 314MB/s]

Upload successful: best_model.pth (81MB)
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100%|██████████| 81.0M/81.0M [00:01<00:00, 74.9MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 12/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 13/300 | loss_G=89.1766 | loss_D=-0.4607 | loss_L1=0.2195 |
val_SSIM=0.9734 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 74.4MB/s]
28%|███████| 22.7M/81.0M [00:00<00:00, 238MB/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 77.3MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 13/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 14/300 | loss_G=91.0004 | loss_D=-0.3229 | loss_L1=0.2157 |
val_SSIM=0.9749 | no improve 1/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 88.7MB/s]
31%|███████| 24.8M/81.0M [00:00<00:00, 258MB/s]

Upload successful: best_model.pth (81MB)

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100%|██████████| 81.0M/81.0M [00:00<00:00, 85.3MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 14/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 15/300 | loss_G=101.5549 | loss_D=-0.2585 | loss_L1=0.2101 |
val_SSIM=0.9762 | no improve 2/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 87.8MB/s]
27%|███████| 21.8M/81.0M [00:00<00:00, 228MB/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 71.5MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 15/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 16/300 | loss_G=107.0323 | loss_D=-0.1504 | loss_L1=0.2035 |
val_SSIM=0.9692 | no improve 3/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 66.0MB/s]
26%|███████| 21.3M/81.0M [00:00<00:00, 210MB/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 66.0MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 16/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 17/300 | loss_G=101.2959 | loss_D=-0.0704 | loss_L1=0.2011 |
val_SSIM=0.9751 | no improve 4/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 81.0M/81.0M [00:00<00:00, 135MB/s]
0%| | 0.00/81.0M [00:00<?, ?B/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 74.9MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 17/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 18/300 | loss_G=127.2995 | loss_D=0.0055 | loss_L1=0.1985 |

val_SSIM=0.9772 | no improve 5/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 76.5MB/s]
28%|███████| 22.5M/81.0M [00:00<00:00, 226MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 74.6MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan3d-normalized>

Cloud: 18/300 -> cminhnguyndsdlds/conditional-gan3d-normalized
Epoch 19/300 | loss_G=104.7737 | loss_D=0.0322 | loss_L1=0.1959 |
val_SSIM=0.9739 | no improve 6/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 81.0M/81.0M [00:01<00:00, 79.9MB/s]
0%| | 0.00/81.0M [00:00<?, ?B/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 69.4MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan3d-normalized>

Cloud: 19/300 -> cminhnguyndsdlds/conditional-gan3d-normalized
Epoch 20/300 | loss_G=107.9143 | loss_D=0.1158 | loss_L1=0.1935 |
val_SSIM=0.9763 | no improve 7/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 76.5MB/s]
27%|███████| 21.8M/81.0M [00:00<00:00, 229MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 80.2MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 20/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 21/300 | loss_G=114.9454 | loss_D=0.1662 | loss_L1=0.1900 |
val_SSIM=0.9782 | no improve 8/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 86.9MB/s]
30%|███████| 24.5M/81.0M [00:00<00:00, 250MB/s]

Upload successful: best_model.pth (81MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 71.3MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 21/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 22/300 | loss_G=121.0012 | loss_D=0.2148 | loss_L1=0.1904 |
val_SSIM=0.9735 | no improve 9/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 84.5MB/s]
30%|███████| 24.4M/81.0M [00:00<00:00, 241MB/s]

Upload successful: best_model.pth (81MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 65.8MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 22/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 23/300 | loss_G=114.9141 | loss_D=0.2979 | loss_L1=0.1886 |
val_SSIM=0.9756 | no improve 10/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version

(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 92.5MB/s]
32%|███████| 26.1M/81.0M [00:00<00:00, 273MB/s]

Upload successful: best_model.pth (81MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 76.5MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan3d-normalized>

Cloud: 23/300 -> cminhnguyndsdlds/conditional-gan3d-normalized
Epoch 24/300 | loss_G=117.3688 | loss_D=0.3452 | loss_L1=0.1854 |
val_SSIM=0.9667 | no improve 11/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 89.2MB/s]
28%|███████| 22.5M/81.0M [00:00<00:00, 236MB/s]

Upload successful: best_model.pth (81MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 72.2MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan3d-normalized>

Cloud: 24/300 -> cminhnguyndsdlds/conditional-gan3d-normalized
Epoch 25/300 | loss_G=113.8259 | loss_D=0.3840 | loss_L1=0.1856 |
val_SSIM=0.9744 | no improve 12/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 70.5MB/s]
25%|███████| 20.3M/81.0M [00:00<00:00, 211MB/s]

Upload successful: best_model.pth (81MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 73.4MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 25/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 26/300 | loss_G=125.6409 | loss_D=0.4753 | loss_L1=0.1820 |
val_SSIM=0.9763 | no improve 13/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 80.9MB/s]
31%|███████| 25.2M/81.0M [00:00<00:00, 263MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 74.4MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 26/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 27/300 | loss_G=119.1787 | loss_D=0.5066 | loss_L1=0.1812 |
val_SSIM=0.9710 | no improve 14/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 71.1MB/s]
32%|███████| 25.6M/81.0M [00:00<00:00, 263MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 68.3MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 27/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 28/300 | loss_G=119.4719 | loss_D=0.5308 | loss_L1=0.1812 |
val_SSIM=0.9729 | no improve 15/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 66.5MB/s]
0%| | 0.00/81.0M [00:00<?, ?B/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:00<00:00, 87.1MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 28/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 29/300 | loss_G=133.2602 | loss_D=0.5650 | loss_L1=0.1785 |
val_SSIM=0.9755 | no improve 16/20
Warning: Looks like you're using an outdated `kaggle` version
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(2.0.2)

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100%|██████████| 81.0M/81.0M [00:01<00:00, 65.8MB/s]
30%|███████ | 24.3M/81.0M [00:00<00:00, 252MB/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 77.1MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 29/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 30/300 | loss_G=114.9004 | loss_D=0.6111 | loss_L1=0.1804 |
val_SSIM=0.9713 | no improve 17/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 81.0M/81.0M [00:01<00:00, 84.6MB/s]
23%|███████ | 18.6M/81.0M [00:00<00:00, 187MB/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 60.5MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 30/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 31/300 | loss_G=116.8031 | loss_D=0.6437 | loss_L1=0.1758 |
val_SSIM=0.9738 | no improve 18/20

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(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 81.0M/81.0M [00:01<00:00, 71.6MB/s]
27%|███████| 21.5M/81.0M [00:00<00:00, 226MB/s]

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100%|██████████| 81.0M/81.0M [00:01<00:00, 81.8MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 31/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 32/300 | loss_G=114.9655 | loss_D=0.7012 | loss_L1=0.1778 |
val_SSIM=0.9704 | no improve 19/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 69.8MB/s]
29%|███████| 23.3M/81.0M [00:00<00:00, 244MB/s]

Upload successful: best_model.pth (81MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 68.3MB/s]

Upload successful: last_checkpoint.pth (81MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan3d-normalized>

Cloud: 32/300 -> cminhnguyndsdssds/conditional-gan3d-normalized
Epoch 33/300 | loss_G=113.0133 | loss_D=0.6947 | loss_L1=0.1746 |
val_SSIM=0.9665 | no improve 20/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 67.1MB/s]
29%|███████| 23.3M/81.0M [00:00<00:00, 244MB/s]

Upload successful: best_model.pth (81MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 81.0M/81.0M [00:01<00:00, 77.8MB/s]

Upload successful: last_checkpoint.pth (81MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan3d-normalized>

Cloud: 33/300 -> cminhnguyndsdsds/conditional-gan3d-normalized
Early stopping tai epoch 33

=== TRAINING HOAN THANH ===

Best val_SSIM: 0.9734

Model lưu tại:

/kaggle/working/conditional_gan3d_normalized/best_model.pth

Push da duoc tích hợp vào training loop (cell tren) – chạy mọi epoch.

print(f'Checkpoints da duoc push len: {KAGGLE_USER}/{DATASET_NAME}')

Checkpoints da duoc push len: cminhnguyndsdsds/conditional-gan3d-normalized